# Run unknown overview

**Total triggers: 4642 (100.00%)**

**Triggers with clusters: 1022 (22.02%)**

**Triggers with BM hits: 1022 (22.02%)**

**Total clusters: 2963 (0.64 per trigger)**

**Avg clusters per trigger (all dets): 0.64**

## Beam configuration

**Date: 2025-09-02**

**Time: 15:34**

**Energy: +1.5**

**Target: Tungsten**

**Cherenkov: H: 14 bar, L: 4 bar**

**Collimator: Â±40**

**Details: File H2C.NP-02.047, beam window opened**

## Avg clusters per trigger (per detector)

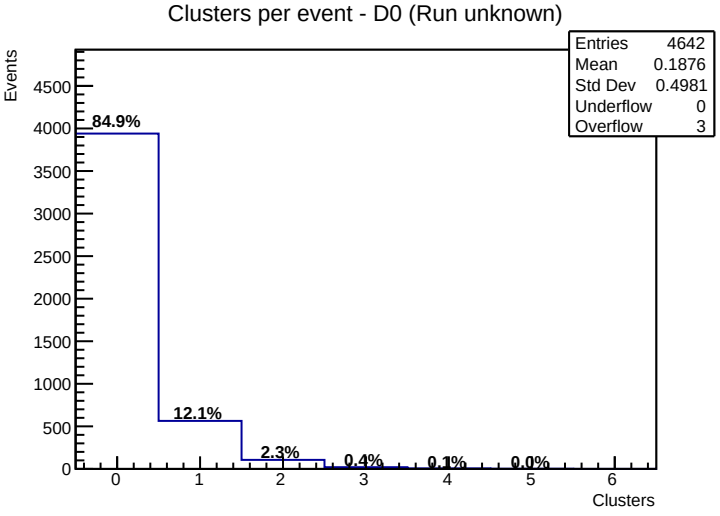
**D0: 0.92 (937/1022)**

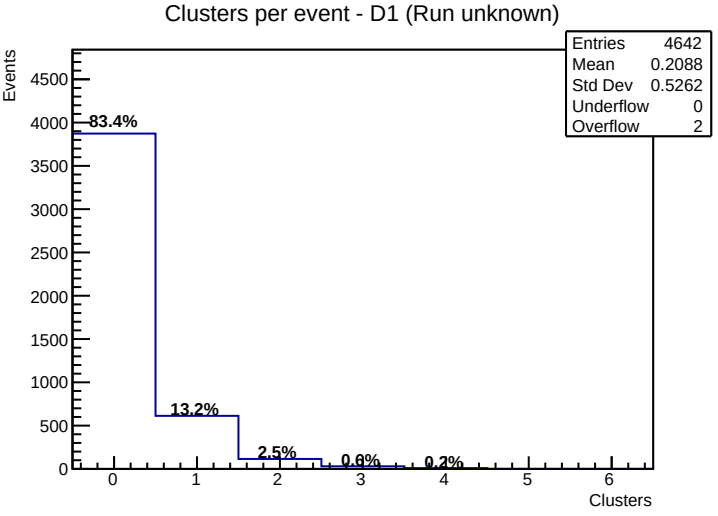
**D1: 0.99 (1012/1022)**

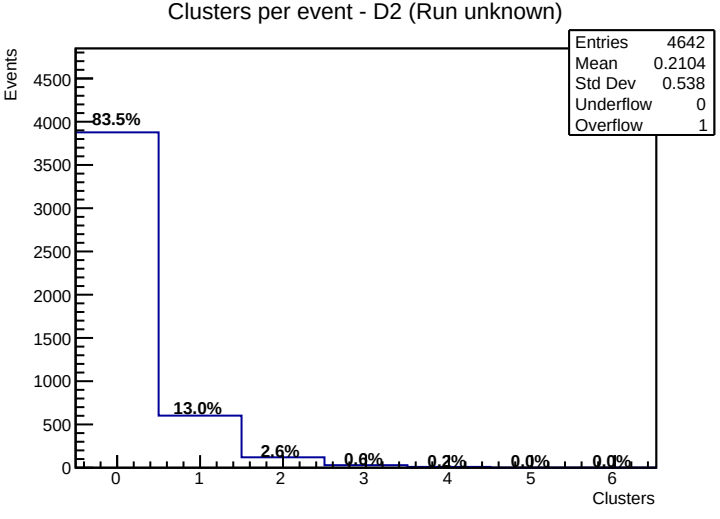
**D2: 0.99 (1014/1022)**

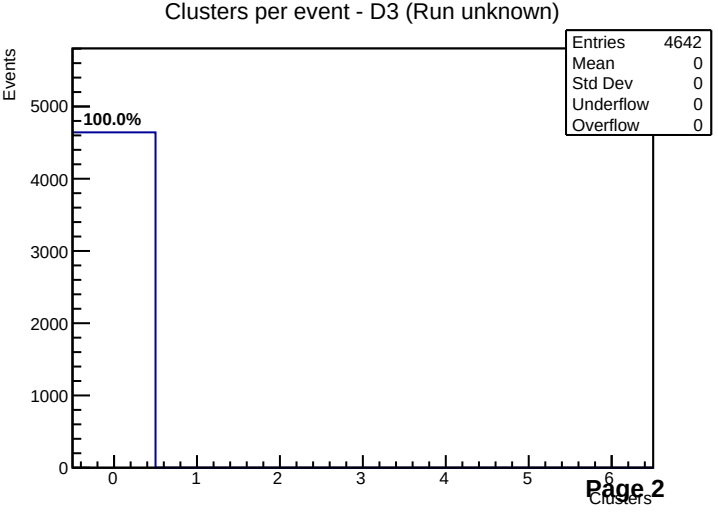
**D3: 0.00 (0/1022)**

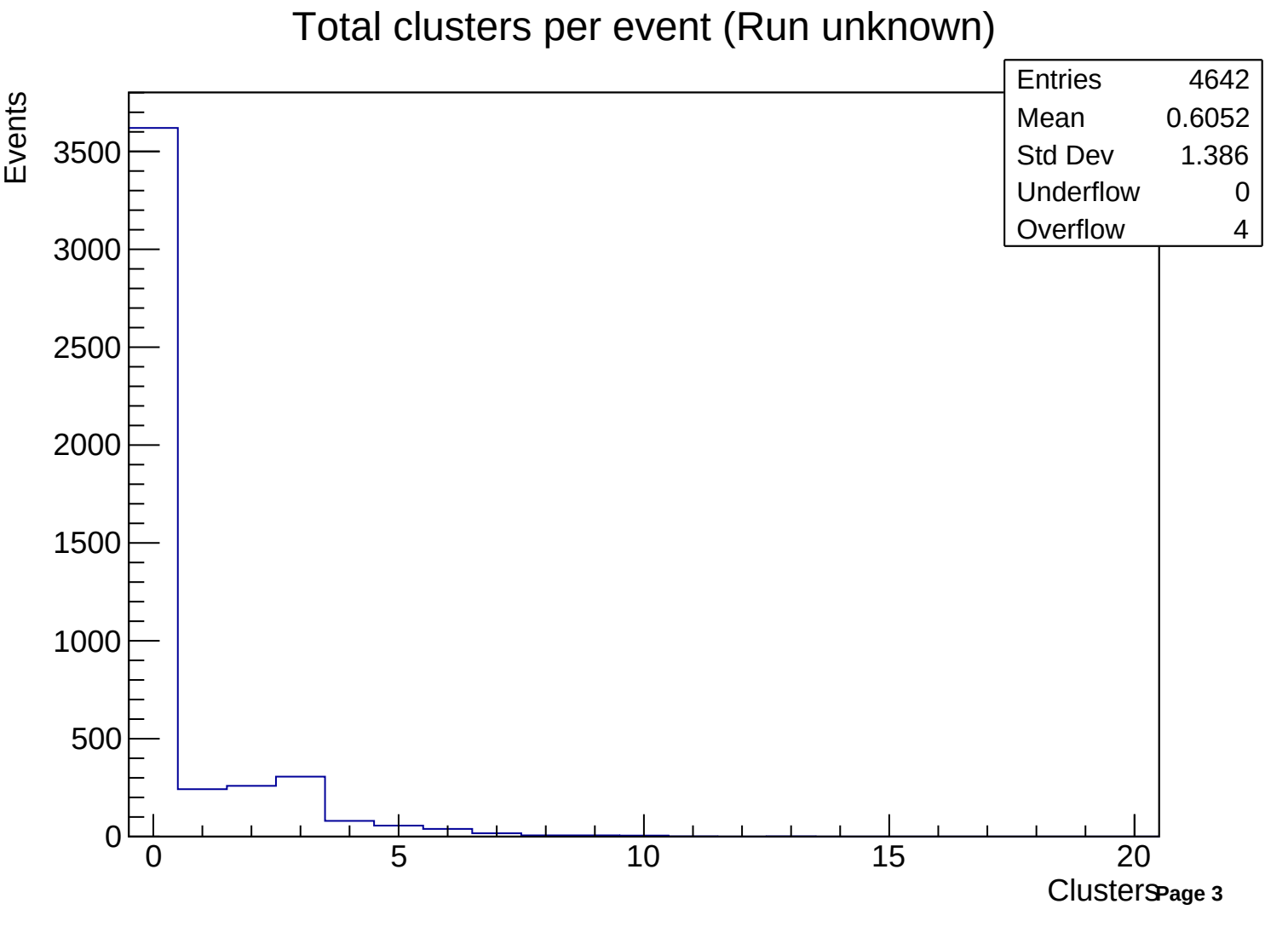
**All averages use triggers with clusters (1022) as denominator.**

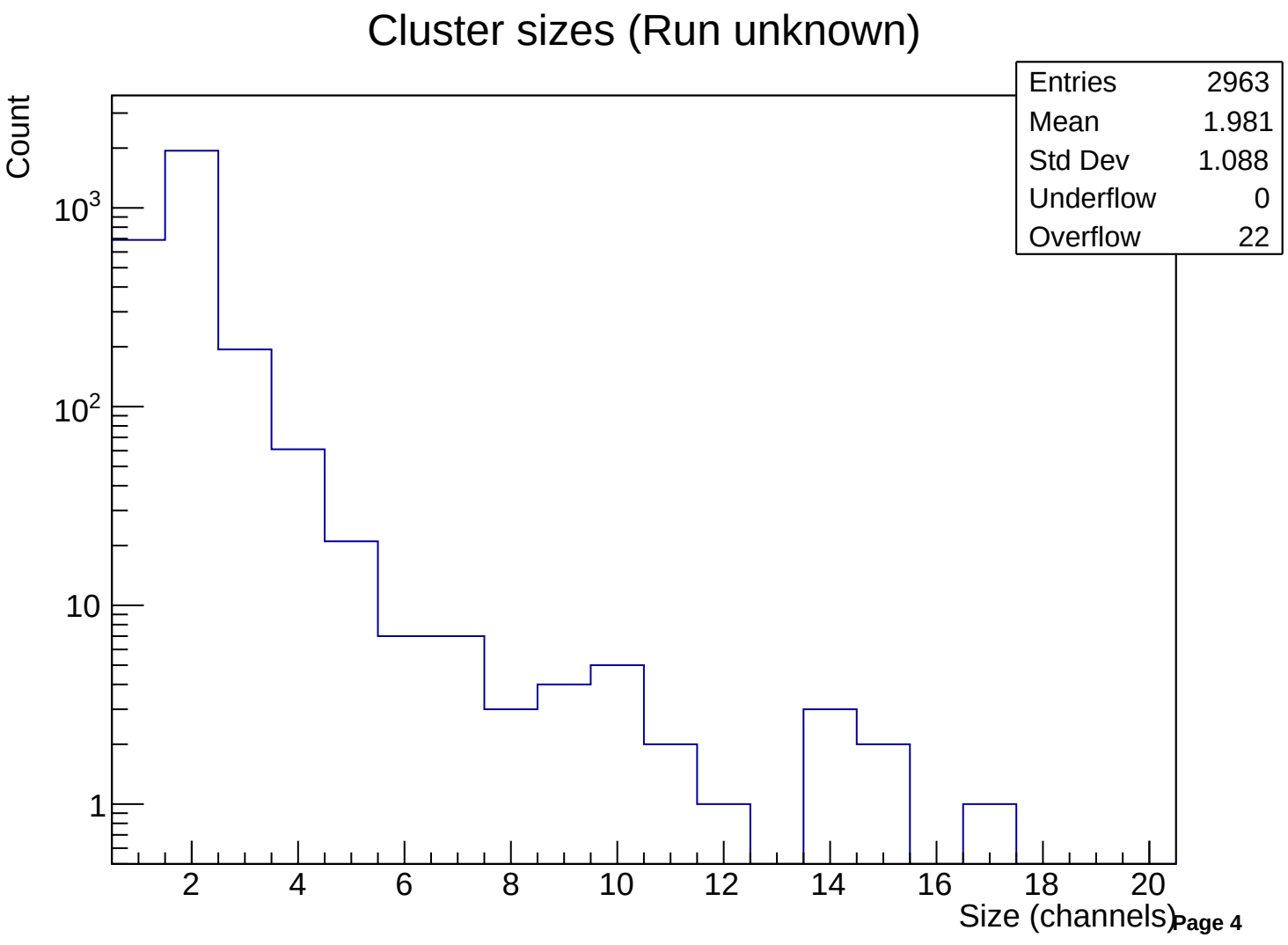

Clusters per event - D0 (Run unknown)
Entries 4642
Mean 0.1876
Std Dev 0.4981
Underflow 0
Overflow 3
Events
Clusters
84.9%
12.1%
2.3%
0.4%
0.1%
0.0%

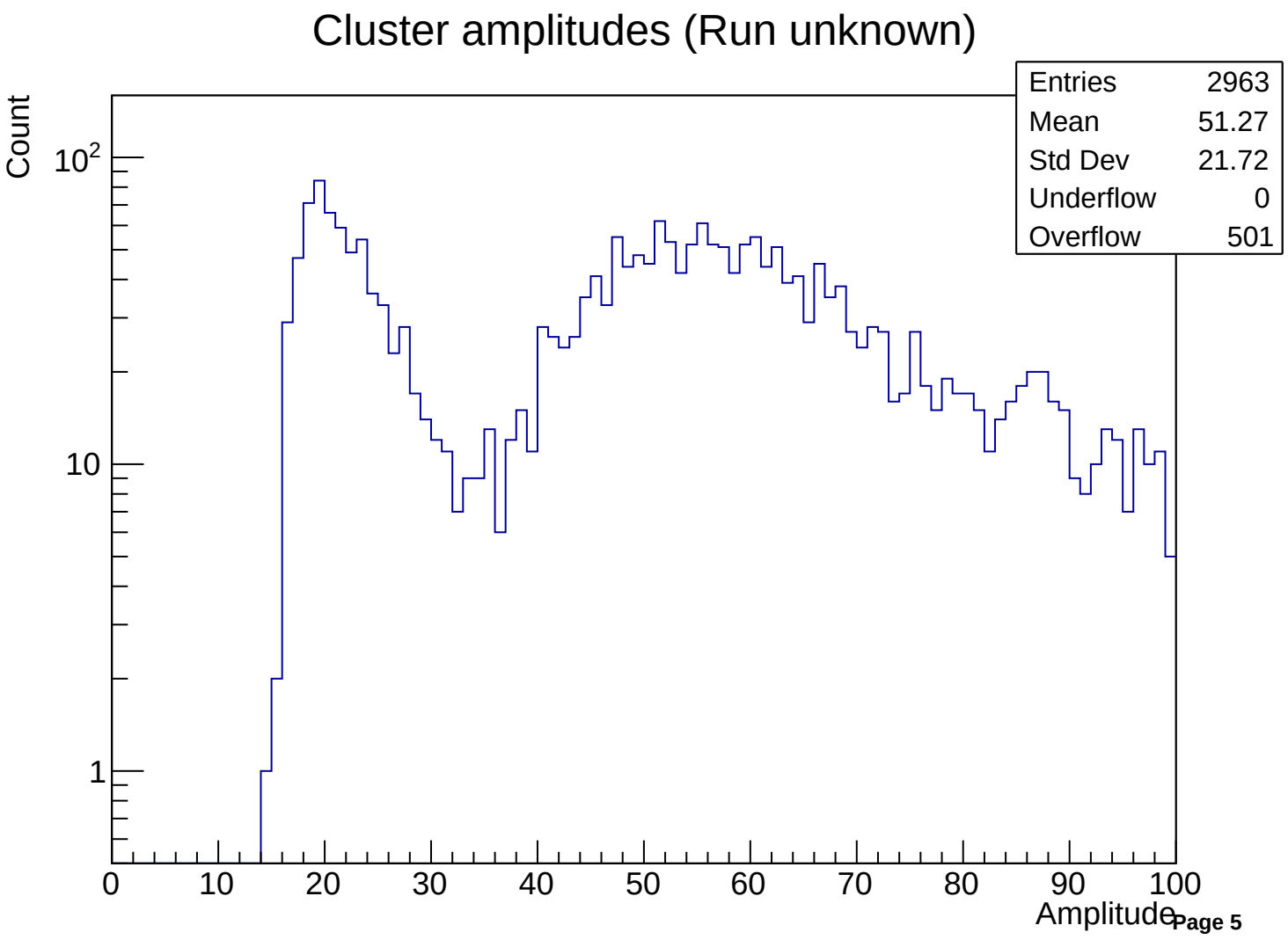

Clusters per event - D1 (Run unknown)
Entries 4642
Mean 0.2088
Std Dev 0.5262
Underflow 0
Overflow 2
Events
Clusters
83.4%
13.2%
2.5%
0.6%
0.2%


Clusters per event - D2 (Run unknown)
Entries 4642
Mean 0.2104
Std Dev 0.538
Underflow 0
Overflow 1
Events
Clusters
83.5%
13.0%
2.6%
0.6%
0.2%
0.0%
0.0%

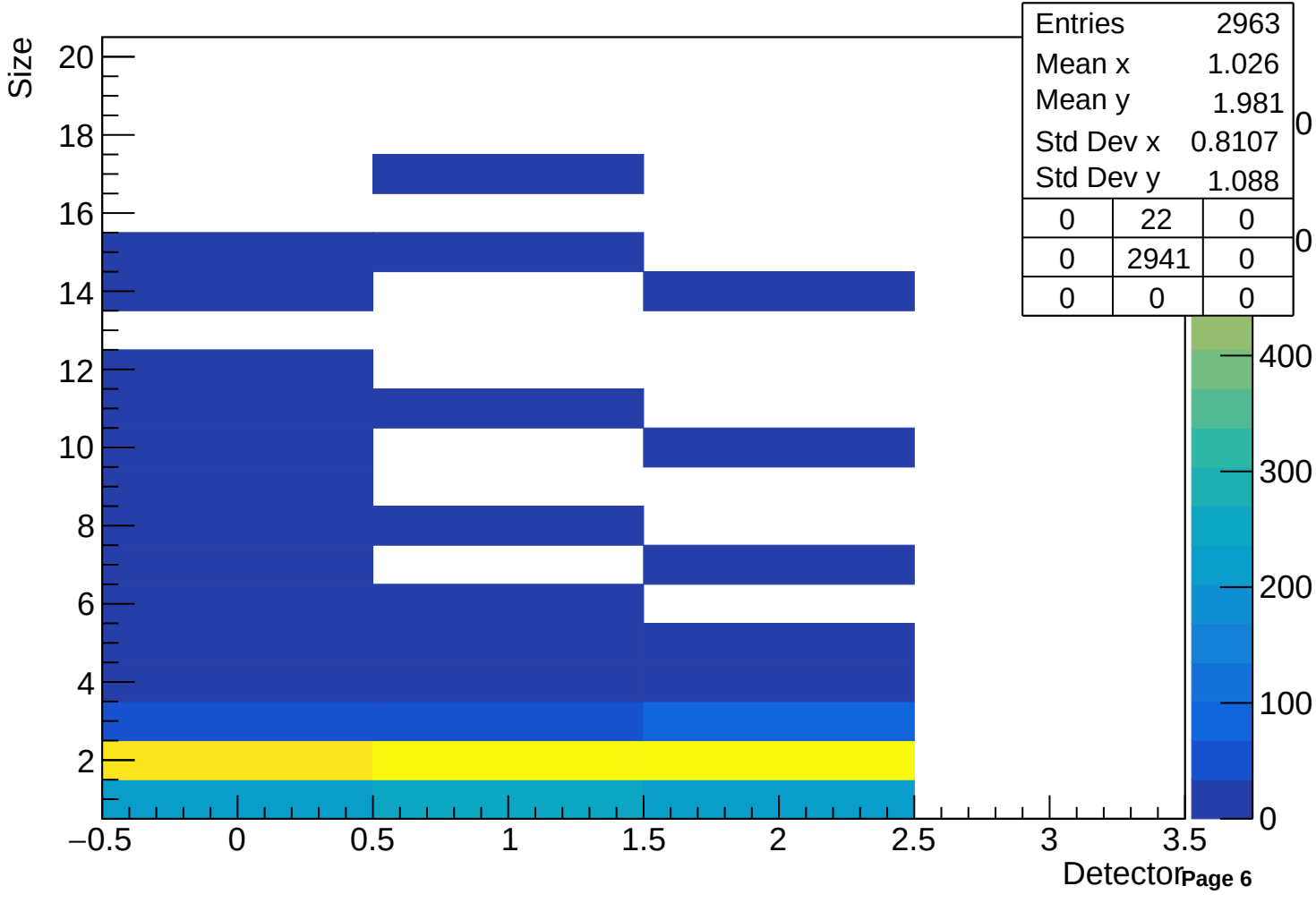

Clusters per event - D3 (Run unknown)
Entries 4642
Mean 0
Std Dev 0
Underflow 0
Overflow 0
Events
Clusters
100.0%


Total clusters per event (Run unknown)
Entries 4642
Mean 0.6052
Std Dev 1.386
Underflow 0
Overflow 4
Events
3500
3000
2500
2000
1500
1000
500
0
0
5
10
15
20
Clusters

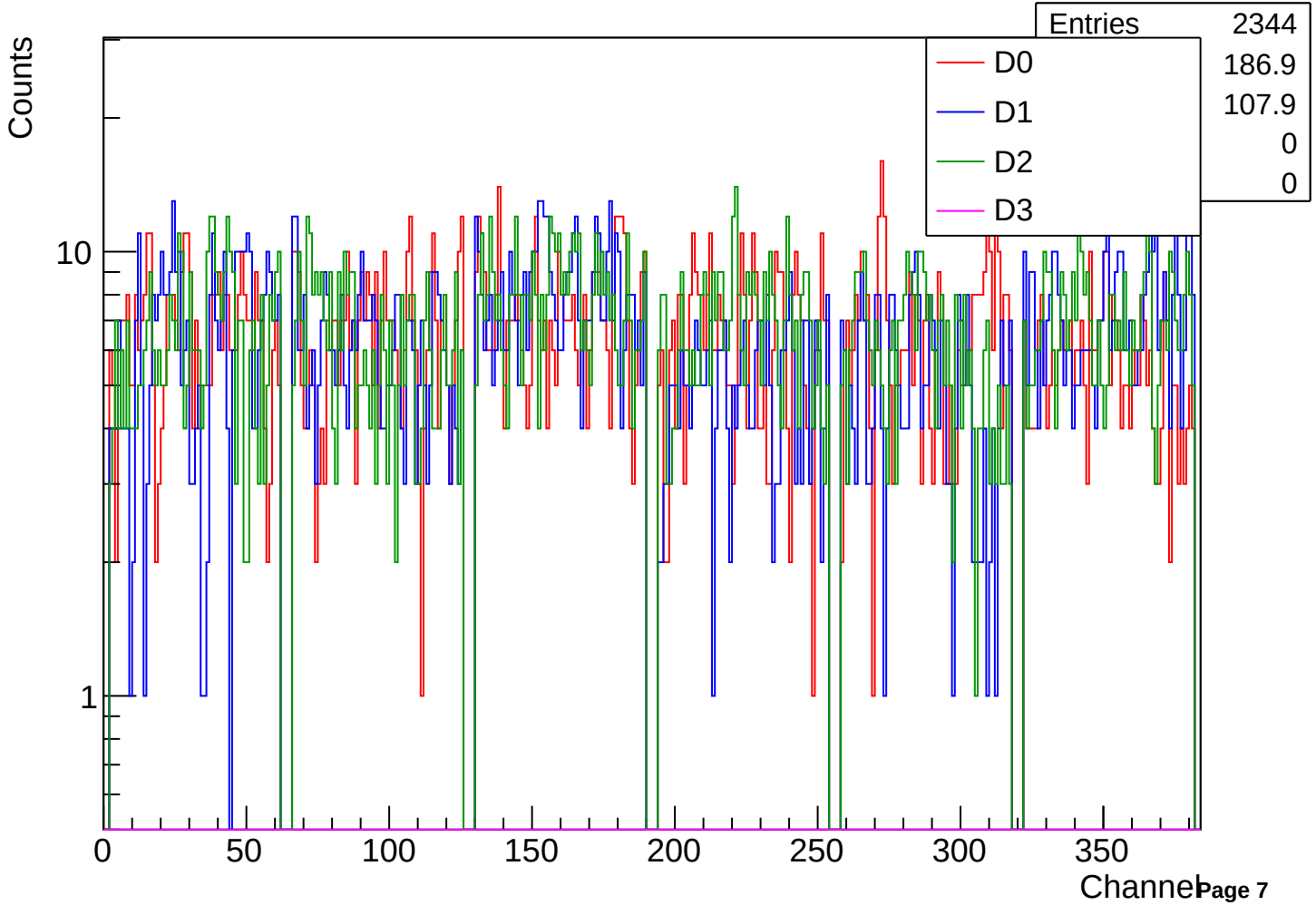

Cluster sizes (Run unknown)
Entries 2963
Mean 1.981
Std Dev 1.088
Underflow 0
Overflow 22
Count
$10^3$
$10^2$
10
1
2
4
6
8
10
12
14
16
18
20
Size (channels)

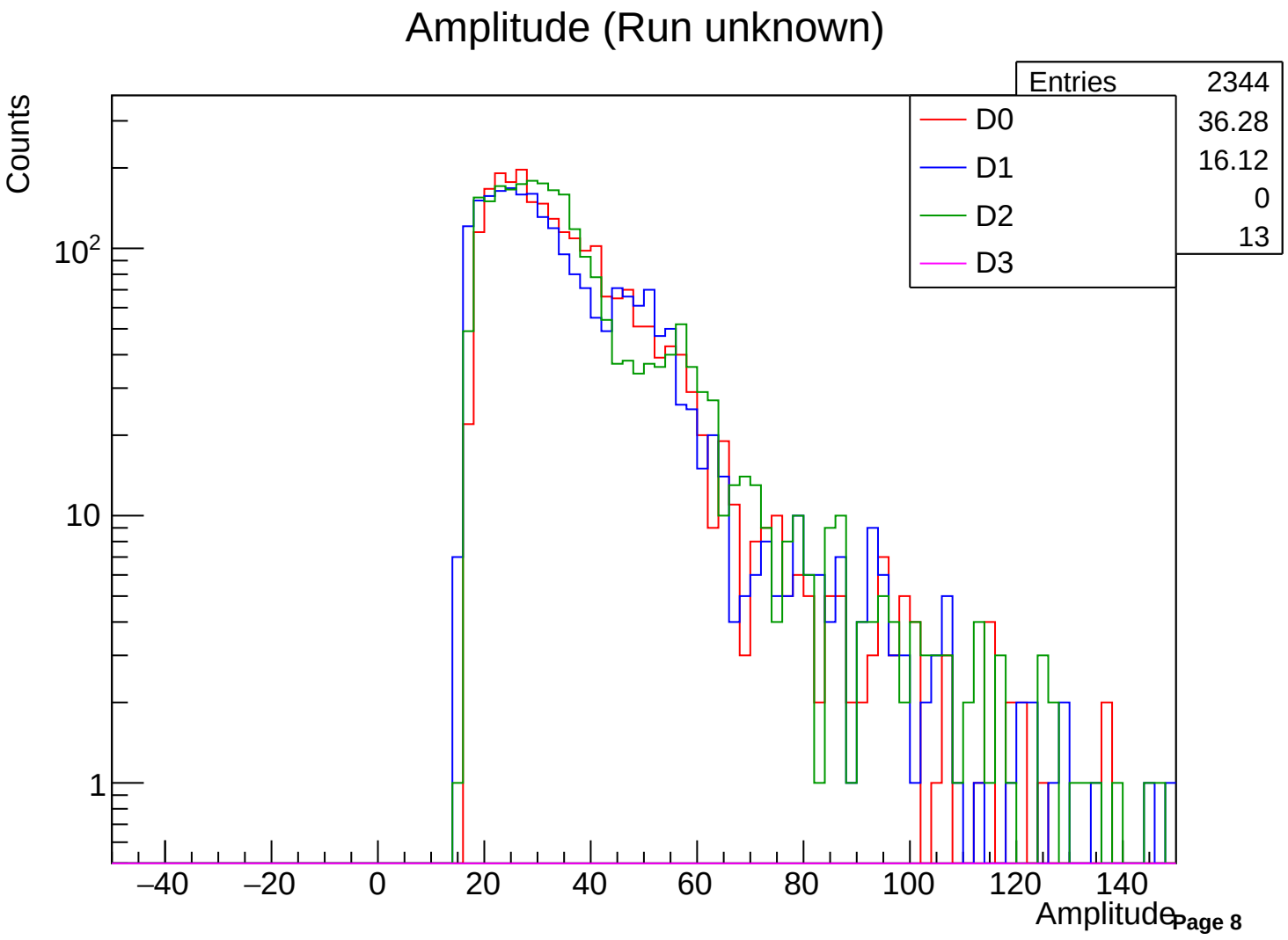

Cluster amplitudes (Run unknown)
Entries 2963
Mean 51.27
Std Dev 21.72
Underflow 0
Overflow 501
Count
$10^2$
10
1
0
10
20
30
40
50
60
70
80
90
100
Amplitude

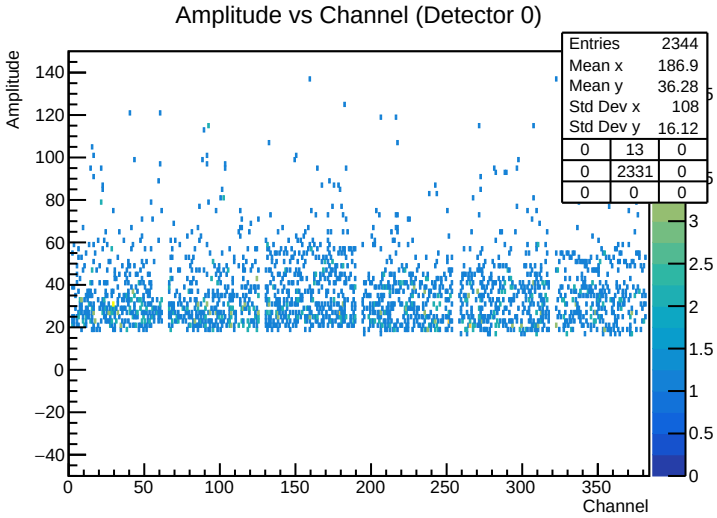
# Cluster size vs detector (Run unknown)

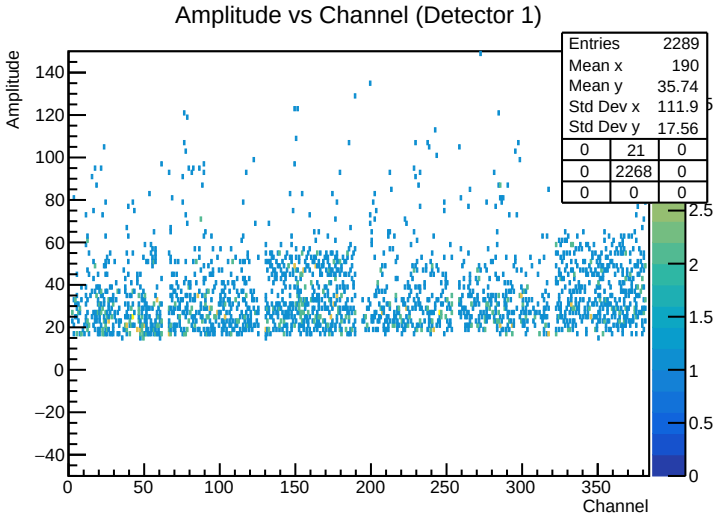


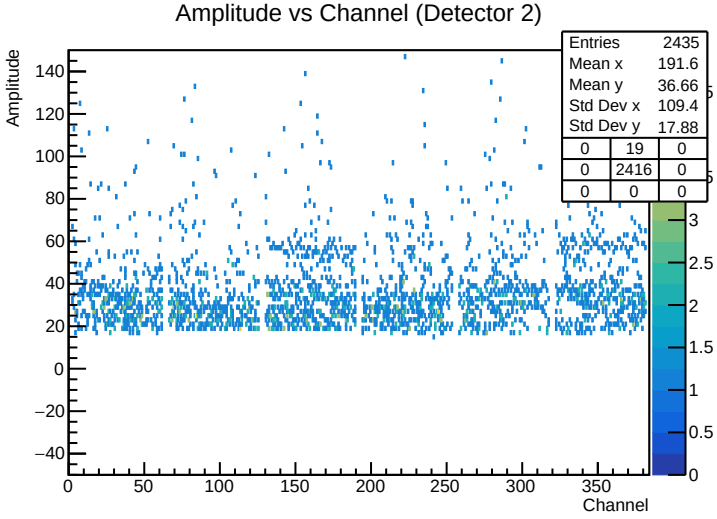
# Firing channels (Run unknown)

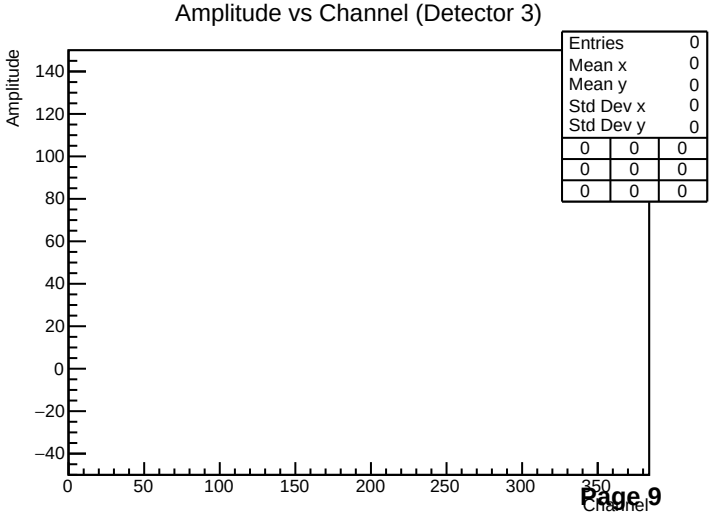



Amplitude (Run unknown)
Entries 2344
36.28
16.12
0
13
D0
D1
D2
D3
Counts
$10^2$
10
1
–40
–20
0
20
40
60
80
100
120
140
Amplitude


Amplitude vs Channel (Detector 0)
Entries 2344
Mean x 186.9
Mean y 36.28
Std Dev x 108
Std Dev y 16.12
0 13 0
0 2331 0
0 0 0
Amplitude
Channel

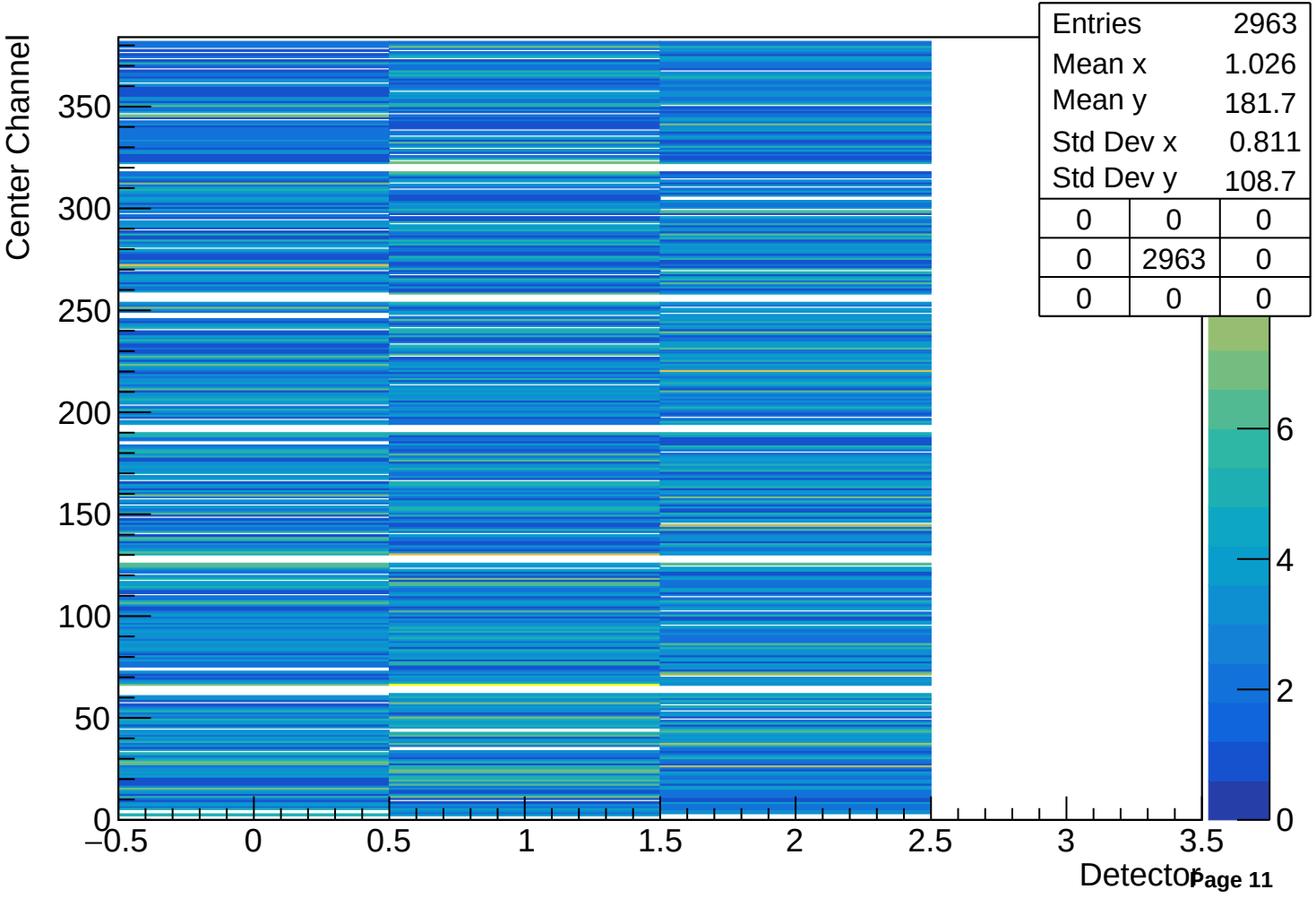

Amplitude vs Channel (Detector 1)
Entries 2289
Mean x 190
Mean y 35.74
Std Dev x 111.9
Std Dev y 17.56
0 21 0
0 2268 0
0 0 0
Amplitude
Channel

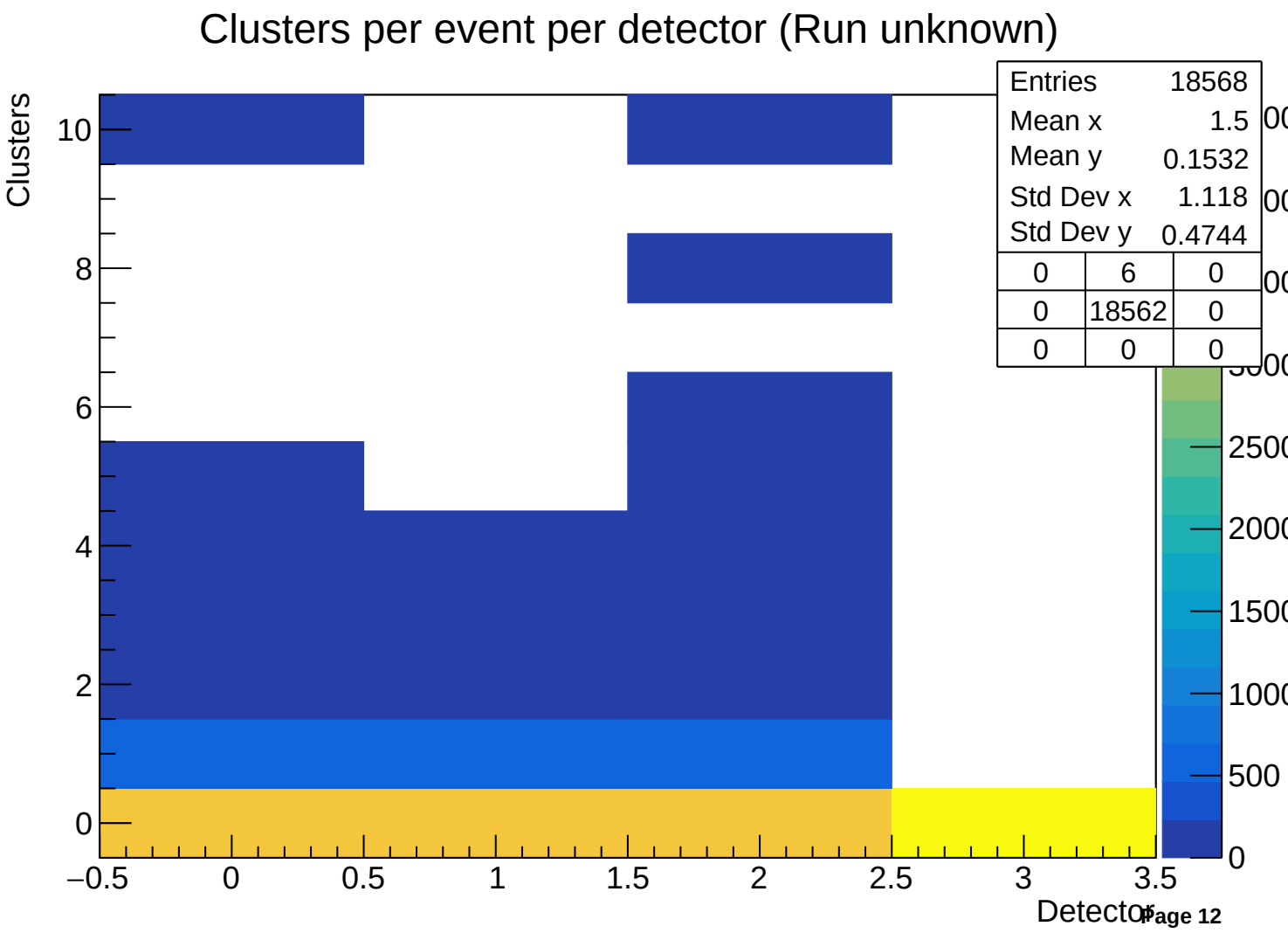

Amplitude vs Channel (Detector 2)
Entries 2435
Mean x 191.6
Mean y 36.66
Std Dev x 109.4
Std Dev y 17.88
0 19 0
0 2416 0
0 0 0
Amplitude
Channel

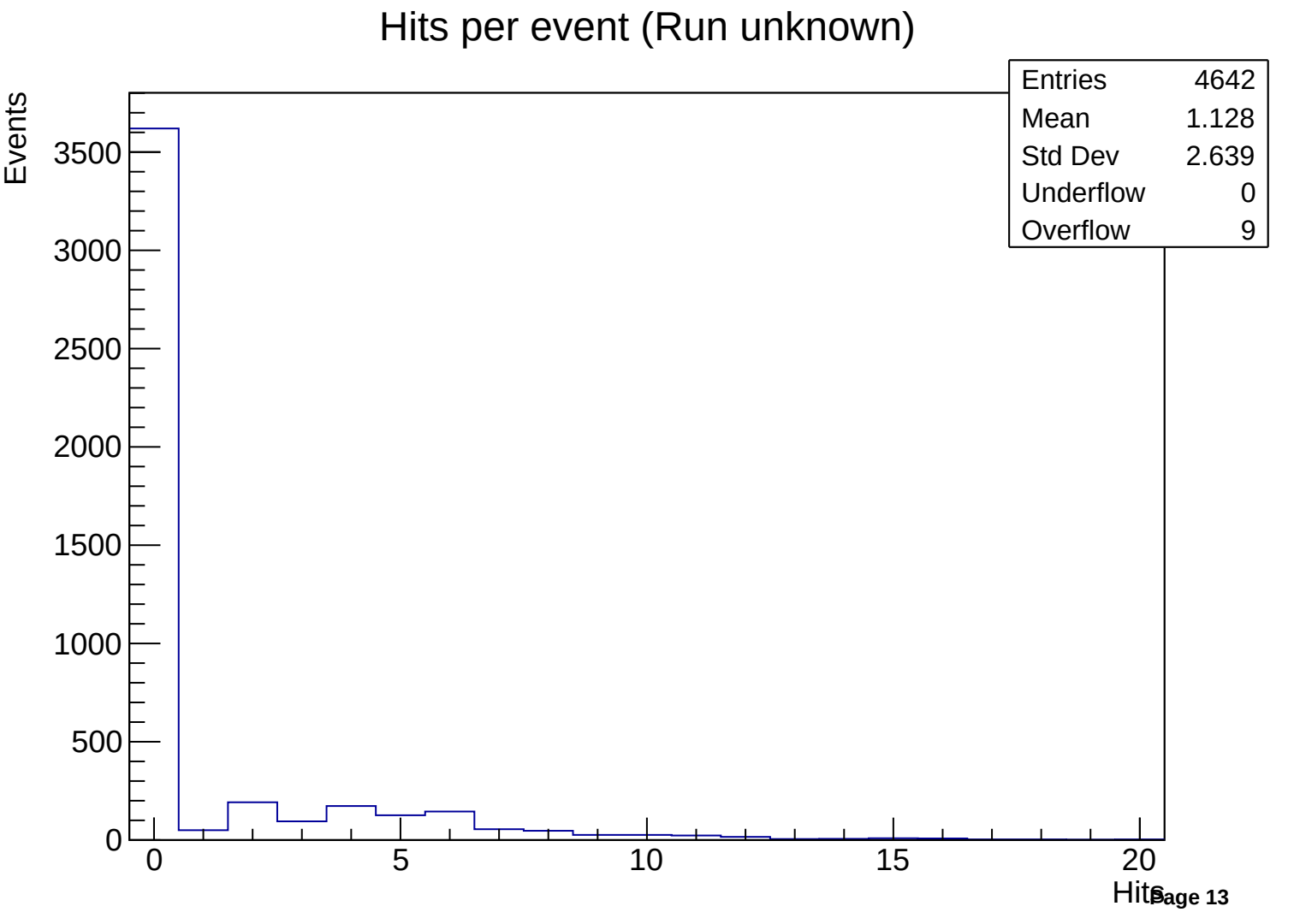

Amplitude vs Channel (Detector 3)
Entries 0
Mean x 0
Mean y 0
Std Dev x 0
Std Dev y 0
0 0 0
0 0 0
0 0 0
Amplitude
Channel

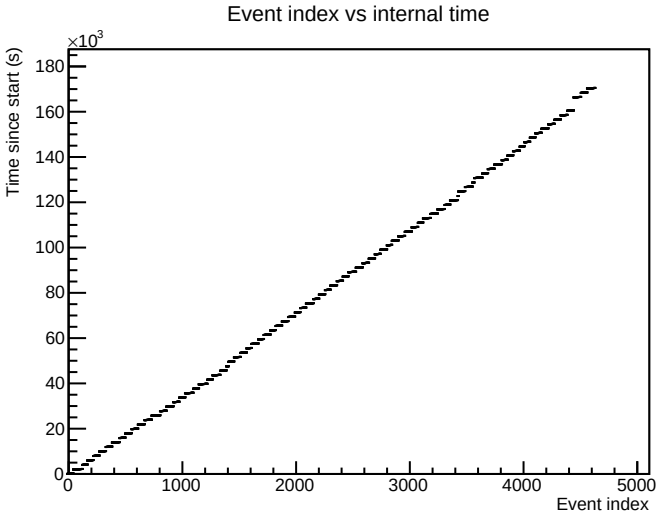
Cluster positions (Run unknown)
Entries 7068
Mean x 1.013
Mean y 189.6
Std Dev x 0.8222
Std Dev y 109.6
0 0 0
0 7068 0
0 0 0
Channel
Detector
0
50
100
150
200
250
300
350
–0.5
0
0.5
1
1.5
2
2.5
3
3.5
0
2
4
6
8
10

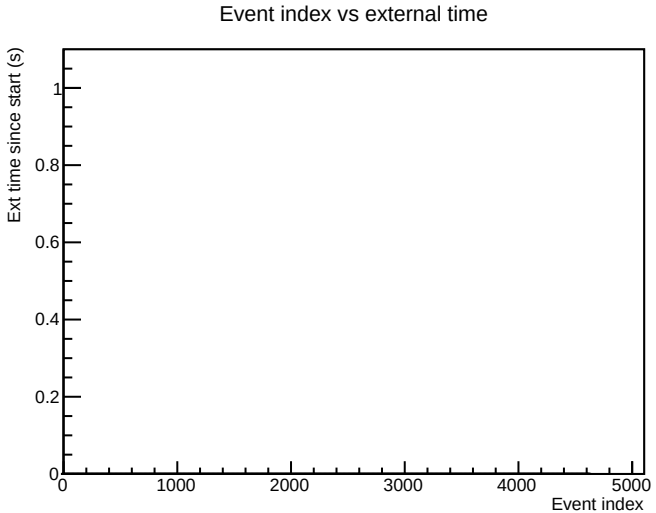
# Cluster center positions (Run unknown)

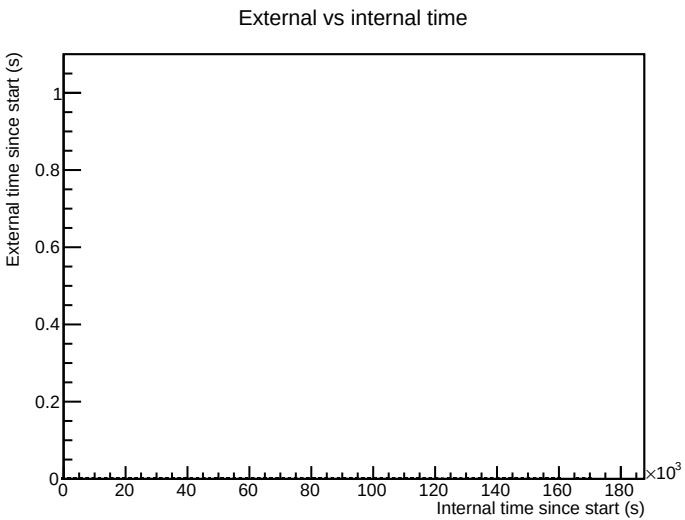


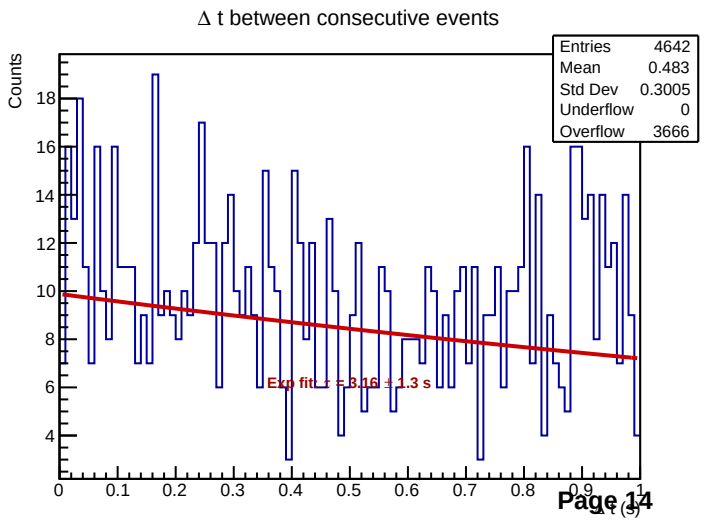

Clusters per event per detector (Run unknown)
Entries 18568
Mean x 1.5
Mean y 0.1532
Std Dev x 1.118
Std Dev y 0.4744
0 6 0
0 18562 0
0 0 0
Clusters
Detector
10
8
6
4
2
0
–0.5
0
0.5
1
1.5
2
2.5
3
3.5
2500
2000
1500
1000
500
0

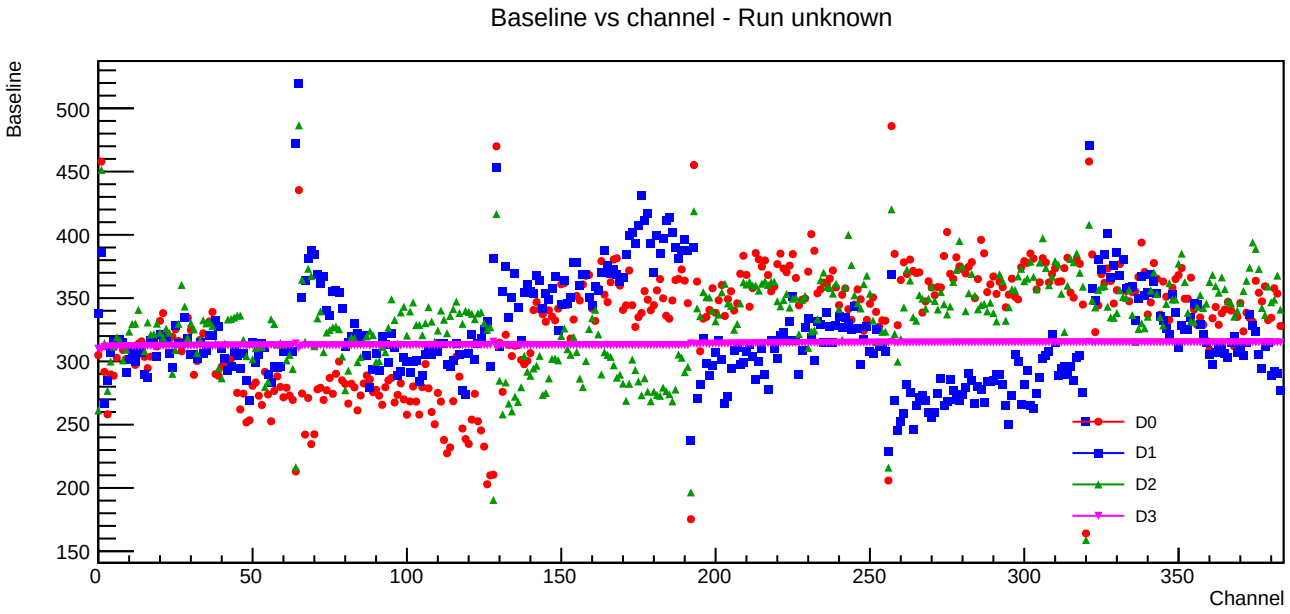

Hits per event (Run unknown)
Entries 4642
Mean 1.128
Std Dev 2.639
Underflow 0
Overflow 9
Events
3500
3000
2500
2000
1500
1000
500
0
0
5
10
15
20
Hits

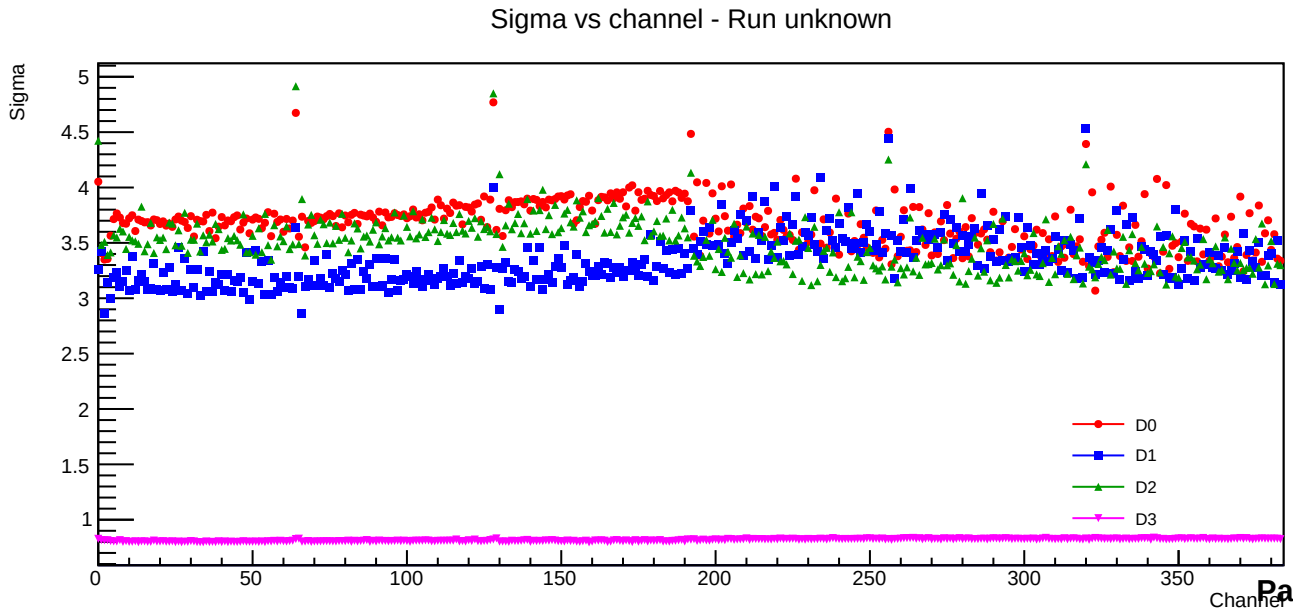

Event index vs internal time
Time since start (s)
×10^3
Event index


Event index vs external time
Ext time since start (s)
Event index


External vs internal time
External time since start (s)
×10^3
Internal time since start (s)


Δ t between consecutive events
Counts
Entries 4642
Mean 0.483
Std Dev 0.3005
Underflow 0
Overflow 3666
Exp fit: τ = 3.16 ± 1.3 s
Δ t (s)


Baseline vs channel - Run unknown
Baseline
Channel
D0
D1
D2
D3


Sigma vs channel - Run unknown
Sigma
Channel
D0
D1
D2
D3